

Taylor Series

Goals For Today

- Be introduced to Taylor series with the simpler version, a Maclaurin series.
- Find the Taylor series for some common functions.
- Apply Taylor series to a physical situation.

Deriving Taylor Series

Integrating and differentiating complicated functions can be very annoying. But integrating and differentiating polynomials is easy! Say we have some arbitrary function, $f(x)$, and some arbitrary polynomial, $a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_nx^n + \cdots$. Is there a way we can write $f(x)$ as some infinite polynomial? Let's find out!

1. We want to find $a_0, a_1, \dots, a_n$ such that

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n + \cdots \quad (1)$$

How could we solve for a_0 ? (Hint: is there a value of x that could make this easier?)

2. How about a_1 ? (Hint: is there an operation we could perform on $f(x)$ that will transform a_1x to a_1 ?)
3. How about a_2 ?
4. How about a_3 ?
5. How can we generalize to find a_n ?
6. Now that we have the formulas for a_n , plug these formulas back in to find $f(x)$. (Hint: use equation 1.)

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7. This is the formula for a Maclaurin series! It's a short jump from here to a Taylor series. Say we only look at the first few terms. For what values of x will $f(x)$ be best approximated by our shortened ("truncated" - we'll see why we can do this in problem 2 of the next section) Taylor series? (Hint: what assumption did we make in problem 1?) How could we change which values of x allow $f(x)$ to be closely approximated? (Hint: how could we modify our approach in problem 1?)

Common Taylor Series

Not all $f(x)$ will have a Taylor series which has a "closed form" format - that is, sometimes you really will need to hand calculate derivatives for the number of terms you need. But some $f(x)$ have a really clean "closed form" solution - a formula which sums up¹ what all the terms of the Taylor series will look like. The problems below have you find the Taylor series for a few really common $f(x)$ that you'll see over and over again.

1. Find the Taylor series for $f(x) = \sin x$. (Assume we're interested in the region near $x = 0$.) Is there a way to write this in closed form format - i.e., using series notation?
2. Why could we truncate the Taylor series in problem 7 of the previous section? Using Desmos, plot the actual function $\sin x$ and then separately $f(x) = a_1x$, and $g(x) = a_1x + a_3x^2$ using what you found in the previous problem. For what values of x are the first few terms of $f(x)$ most accurately approximating $\sin x$? How about $g(x)$? Add functions that add more terms of the Taylor series. As you get to higher order terms (note that order refers to the power that x is raised to in the term; i.e. a_1x is the first order term, etc), is the approximation changing significantly?

¹I will not apologize for this pun.

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3. (Optional.) Find the Taylor series for $f(x) = \cos x$, e^x , and $f(x) = (x+1)^p$. (Assume we're interested in the region near $x = 0$.) Does this last one look familiar? What do the coefficients work out to? (Look up Pascal's triangle, the binomial theorem, and the idea of a combination/permutation if you want to fall down a cool rabbit hole!)
4. (Optional, but especially recommended.) A really common approximation is the "small angle approximation", where $\sin \theta \approx \theta$ for $\theta \ll 1$. If $\theta \ll 1$, what can we approximate $\cos \theta$ as? How about $\tan \theta$? Find one or two physics formulas this simplifies.

Manipulating Taylor Series

Now that we have derived the Taylor series for some common functions, we can see how it would be tedious to find Taylor series for more complex expressions if we had to take a bunch of derivatives every time. But there's a couple of ways we can use existing Taylor series to make new ones.

1. Take the first couple of derivatives of $f(x) = \sin(x) + \cos(x)$. What does this tell you about the Taylor series of $f(x)$?
2. How could you construct $\sin(2x)$? e^{x^2} ?
3. The series for $g(x) = \frac{1}{1-x}$ is $1 + x + x^2 + x^3 + \cdots + x^n$. How is $g(x)$ related to $-\frac{1}{(1-x)^2}$? How could you change the Taylor series for $g(x)$ to get $-\frac{1}{(1-x)^2}$?

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Using Taylor Series

We have Taylor series - let's examine how to use them in a physical, practical context.²

The first issue is what we “expand about”. We discussed in the first section (problem 7) how our formula was most accurate when x was close to zero (or, when we rewrote our formula, when x was close to a). (See also problem 2 from the previous section.) But what does this mean when x has a dimension/unit, as it often does in physics? For instance, say our $f(x)$ has an x which is a distance. When we say that we're interested in x being near zero, it matters whether x is in units of parsecs or centimeters! Because of this, we need x to be dimensionless, so we can easily compare it to 0.

1. (Optional.) Let's say we are dealing with projectile motion with air resistance. Say that air resistance can be modelled as a force $\vec{F}_{drag} = -b\vec{v}$ (i.e., that acts directly opposite to the direction the object is moving). Draw the free body diagram for this situation.
2. (Optional.) Using the free body diagram from the previous problem, set up the equations for motion in the x and y directions.
3. Next week, we'll go over how to solve differential equations like the ones set up in the previous problem - but for now, we'll skip past this. It turns out the equation for the trajectory is

$$y(x) = \frac{gm^2}{b^2} \ln \left(1 - \frac{xb}{mv_{x,0}} \right) + \frac{v_{y,0} + \frac{mg}{b}}{v_{x,0}} x \quad (2)$$

Where $v_{x,0}$ and $v_{y,0}$ are the initial velocities in the x and y directions, respectively. This looks nasty - but it should look like our regular equations for parabolic motion in a

²Parts of this adapted from https://physicscourses.colorado.edu/phys2210/phys2210_fa20/lecture/lec10-taylor-series/.

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vacuum as $b \rightarrow 0$. Let's check that it does! First, why can't we simply Taylor expand about b ? (Hint: what units does b have?)

4. Define some dimensionless quantity ϵ that we *can* Taylor expand about. (Hints: how can we get the units of b to cancel? How can we get a relationship where $\epsilon \ll 1$? What part of equation 2 is the “grossest”? And how can we get rid of all the b 's in that part of equation 2?)
5. Taylor expand the first term of equation 2 about the ϵ you found in the previous problem and simplify. Confirm that this matches the equation for projectile motion in a vacuum,³

$$y_{vac}(x) = \frac{v_{y,0}}{v_{x,0}}x - \frac{x^2 g}{2v_{x,0}^2} \quad (3)$$

³Last week we talked about ways to check your answer. This is us checking a more complicated limiting case!

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Reflection

Take some time to think through these questions. Part of purposeful practice is thinking through what your practice has taught you, how you will use it in the future, and what you might do differently next time, so we want to give you the space to do this.

1. What did you learn from these problems? How did you know to use the techniques you used? How might you apply this to a future problem?
2. How might you connect this to past material you've learned?
3. What questions do you have?